

YUVIKA MEHTA

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Professional Summary

Computer Science undergraduate with internship experience in **machine learning**, **generative AI**, and **data automation** across industry and research settings. Skilled in building **end-to-end ML pipelines**, **NLP-powered applications**, and **full-stack tools** with strong quantitative, analytical, and collaborative abilities.

Education

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science and Engineering — CGPA: 9.86/10

Jun 2023 – May 2027

Kattankulathur, Tamil Nadu

- Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Machine Learning, Probability and Queuing Theory

Experience

Machine Learning Intern

Dec 2025 – Mar 2026

JA Assure

Remote

- Built a **multilingual conversational AI assistant** supporting **7 languages** with voice/text interaction using **FastAPI**, **Whisper STT**, and **Edge TTS**, powered by a local **LLaMA 3.2 LLM** via Ollama.
- Implemented a **RAG pipeline** using **Sentence-BERT** embeddings and **FAISS** vector search to retrieve and ground responses from insurance policy documents.

Machine Learning Intern

Jun 2025 – Jul 2025

Indian Institute of Technology Kharagpur

Kharagpur, West Bengal

- Developed **automated ML pipelines** on **30,000+ manufacturing samples** for model training, evaluation, and comparison (Random Forest, XGBoost, ANN, LSTM), achieving **$R^2 = 0.99$** .
- Built an interactive **Streamlit dashboard** for performance visualization, experiment tracking, and automated reporting.

Generative AI Intern

Dec 2024 – Jan 2025

Tata Steel Downstream Products Limited

Kolkata, West Bengal

- Engineered an **LLM-powered automation framework** using **Python** and **NLP** to streamline procurement workflows, reducing manual review time by **20%** across 5 departments.
- Developed **ProcurAIytics**, an AI chatbot using **LangChain** that automated **SQL query generation** from natural language, enabling **3× faster data retrieval** with **40%** efficiency gain.

Technical Projects

CoalMeth – Methane Detection — Python, Scikit-learn, Google Earth Engine, Rasterio — [GitHub](#) Dec 2025 – Mar 2026

- Developed a **satellite-based methane monitoring system** for **8 Indian coal mines**, predicting **360.8 Gg CH₄/year** emissions with **$R^2 = 0.977$** , validated against TROPOMI observations (ratio = 1.04).
- Built an **ML classification pipeline** achieving **91.7% accuracy** and **94.1% F1-score** for coal mine detection using **Sentinel-2** spectral features with LOO-CV, integrating multi-satellite data into a reproducible climate monitoring framework.

OncoVision – Breast Cancer Prediction — Python, XGBoost, Random Forest, CNN — [GitHub](#)

Nov 2025

- Developed an **end-to-end ML pipeline** integrating ensemble models (XGBoost, Random Forest) with **CNN-based image classification**, achieving **97%+ accuracy** through systematic experimentation.
- Implemented **Explainable AI** techniques (**SHAP**, **LIME**) for model interpretability and deployed a real-time **Streamlit** web application for interactive visualization.

PerzonAI – AI Marketing Tool — Next.js, FastAPI, Google Gemini API, PostgreSQL — [GitHub](#)

Aug 2025

- Built a **full-stack application** with **FastAPI** backend and **Next.js** frontend, integrating **Google Gemini API** for AI-powered persona generation and conversation simulation.
- Designed **PostgreSQL** schema with **automated data export pipelines** and **RESTful APIs**, improving marketing campaign efficiency by **60%**.

Technical Skills

Languages: Python, SQL, C++, Java, JavaScript

ML / AI: Scikit-learn, TensorFlow, PyTorch, LangChain, Hugging Face Transformers, NLP, LLMs, RAG

Data & Analytics: Pandas, NumPy, SciPy, EDA, statistical modeling, hypothesis testing, feature engineering

Backend & Frontend: FastAPI, PostgreSQL, MongoDB, REST APIs, Next.js, Streamlit

Tools: Git, VS Code, Jupyter Notebook, Google Colab, Microsoft Excel

Achievements and Certifications

Academic Excellence: Recognized in the **Top 2% of 2,000+ students** in the CS Department (2024–25).

Hackathon: Won **1st Prize** at AgentForce AI Hackathon (Your Engineers) for PerzonAI.

Certifications: Certified Generative AI Developer (SAP), Certified System Administrator (ServiceNow), Programming in Java (NPTEL)